

E-COMMERCE DATASET CLEANING AND VISUALIZATION DOCUMENTATION

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Dataset: Online Retail Transaction Records

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OBJECTIVE

To explore the e-commerce transaction data, perform data cleaning to handle missing and invalid values, create new features for analysis, and uncover insights through exploratory data analysis and visualization. The aim is to understand sales performance, customer behavior, product performance, and geographic revenue patterns.

DATASET OVERVIEW

The e-commerce dataset contains online retail transaction records from 1 Dec 2010 to 9 Dec 2011. Each row represents a product transaction and includes details such as InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country. The original dataset had 541,909 rows and 8 columns.

LIBRARIES USED: pandas, matplotlib.pyplot, seaborn

DATA CLEANING PROCESS

The dataset was first loaded into Python using pandas. The file did not load properly with the default encoding, so it was loaded using encoding='latin1'. After loading the data, I checked the structure of the dataset, viewed the first few rows, checked the number of rows and columns, reviewed the data types, and identified missing values.

STEPS	ROWS	COLUMNS	ACTION TAKEN
Raw Dataset (Before cleaning)	It contained 541,909	It contained 8	Loaded Online Retail .csv with latin1 encoding.
Missing Values	541,909	8	Filled missing Description and CustomerID with "Unknown"
Invalid Data Removal	530,104	8	Removed rows where UnitPrice <= 0 or Quantity <= 0
Feature Engineering	530,104	13	Created Revenue, Year, Month, Day, MonthName

Final Clean Dataset	530,104	13	Dataset used for EDA and visualization
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NEW COLUMNS ADDED

Revenue = Quantity * UnitPrice.

Year, Month, Day, MonthName = Extracted from InvoiceDate for trend analysis

NOTE: No rows were dropped due to missing values. Only ~2.2% of rows were removed due to invalid transactions.

METRIC	VALUE TOTAL
Revenue	£9,747,747.93
Unique Customers	4,372
Unique Products	4,070
Countries Served	38
Avg Order Value	£376.36

VISUALIZATIONS CREATED

The following visualizations were created for the e-commerce dataset:

1. Top 10 products by quantity sold - to see traffic-driving products
2. Top 10 countries by total revenue - to understand geographic performance
3. Monthly revenue trend - to identify seasonal patterns using MonthName
4. Top 10 Most Purchased Products by Invoice Count- to find the best-selling product
5. Top 5 customer by total spend- to identify high-value customers

Each visualization was used to better understand sales performance, customer contribution, product performance, and country-level revenue patterns.

KEY FINDINGS

1. Top-selling products are dominated by a few high-demand decorative items, with one product significantly leading in quantity sold, product like WHITE HANGING HEART T-LIGHT HOLDER ranked 1st, followed by REGENCY CAKESTAND 3 TIER and JUMBO BAG RED RETROSPOT. This indicates strong customer preference patterns, creating opportunities for better inventory planning and targeted marketing strategies.

2. The United Kingdom overwhelmingly dominates revenue, contributing far more than all other countries combined. Other countries like Netherlands, EIRE, and Germany contribute relatively small and similar amounts, indicating low market penetration outside the UK.

3. Sales fluctuate in the first half of the year, then show a steady increase from September to November, peaking in November. There is a sharp drop in December, likely due to incomplete data or end-of-period recording.

4. A few affordable, gift style products appear on the most customer invoices, making them the main traffic drivers for business. The no 1 top purchased item is WHITE HANGING HEART T-LIGHT HOLDER AND REGENCY CAKESTAND 3 TIERS. This reflects strong customer demand.

5. High-value customers contribute disproportionately to total revenue, emphasizing the importance of customer retention strategies. The CustomerID 14646.0 is the most spender.

RECOMMENDATIONS

1. Ensuring availability for best -seller especially the top 3 to avoid stockout and lost sales. Use top products in promotions to drive revenue.
2. Capitalized on peak months (Sept-Nov) that is increase inventory, market campaign during high demand period.
3. Expand marketing and sales efforts in other countries to reduce over-reliance on the United Kingdom, this will help boost revenue in underperforming countries.